# Prativimb

*A patient twin that answers what if, not just what next.*

---

Hybrid digital twin for Type 2 diabetes and cardiovascular risk, built for Indian patients.

Happiest Health hackathon. Digital Twin track, Phase 1: prototype and code.


<TeamName>, <CollegeName> / <Incubator>

<Member 1, role>  <Member 2, role>  <Member 3, role>  <Member 4, role>  <team email>


Prativimb is Sanskrit for reflection. Patient data shown is synthetic; real data validation uses open access CGMacros.

WHY NOW

# The problem

- India has about 101 million adults with diabetes and 136 million with prediabetes (ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023).
- Most see a doctor every three months. Glucose, sleep and heart rhythm change every five minutes.
- CGMs and watches now record that stream, but they show what already happened.
- Risk scores say who is at risk. They do not say what to change today, or how sure they are.

What a clinician needs at the bedside

- Which patients will go out of range in the next two hours?
- Why, in words, not feature names?
- What would help this patient: a walk, a different grain, an earlier dinner?
- Who is drifting towards a cardiac event this week?

OUR ANGLE

# Why a twin, and not another classifier

| | Risk classifier | Prativimb twin |
|---|---|---|
| Question | What happens next? | What happens next, and what if we change something? |
| Model | Learned from data only | Physiology (ODE) plus a learned residual |
| Personal | Population average | Parameters fitted to each patient's first week |
| Out of range inputs | Unreliable | Physics extrapolates plausibly |
| Uncertainty | A score | Calibrated interval with stated coverage |
| Reasons | Feature importances | mg/dL contributions, one plain sentence |

*Prativimb is a reflection of the patient you can question.*

METHOD

# Our approach in four ideas

## 1. Hybrid twin

Bergman minimal model extended with meals, exercise, sleep debt, circadian rhythm, night shifts and metformin. EHR sets the prior, the first week of wearables fits 8 personal parameters, LightGBM learns what the physics misses.

## 2. Counterfactuals

Re-simulate the next six hours after a walk, a millet swap, a shifted dinner, a missed dose, more sleep or a night shift. Shows peak and time above range for both.

## 3. Honest uncertainty

Conformalised quantile regression gives 90% intervals that reach 89.2% coverage at 60 min on unseen patients. Alerts fire only when the lower bound crosses 180.

## 4. India first

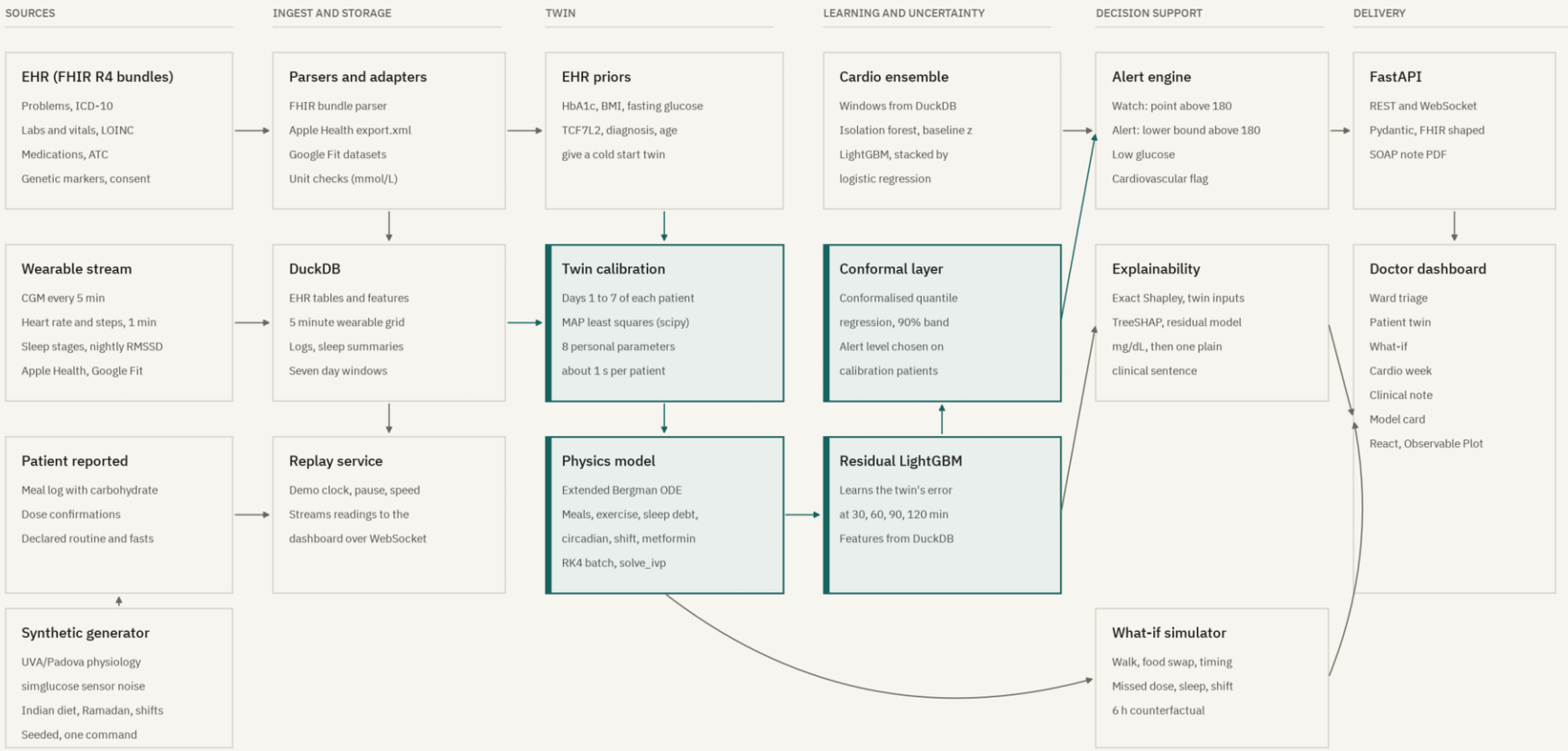
Rice, roti, idli and iftar meals with their absorption speeds; Ramadan, Shivaratri, Ekadashi and Holi in the calendar; night shift workers; cardiovascular risk at lower BMI.

SYSTEM

# Architecture

## Prativimb architecture

Hybrid patient twin: mechanistic physiology, learned residual, calibrated uncertainty, plain language reasons.

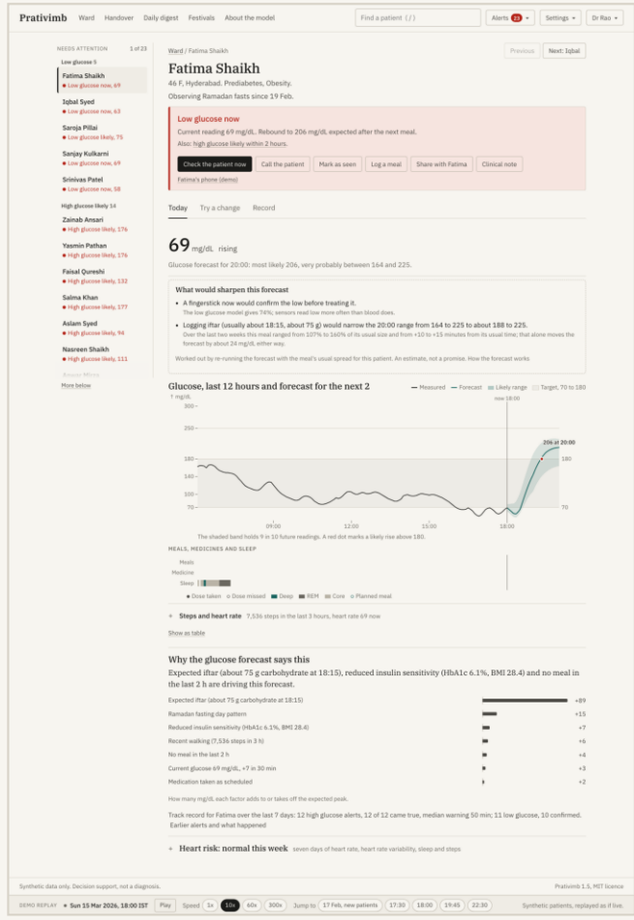



Hybrid core: physics gives plausibility and extrapolation, the residual model gives accuracy, conformal gives honest uncertainty.

Synthetic data only. Decision support, not a diagnosis.

# Data: synthetic, Indian, reproducible

- 200 fictional patients, 30 days each (14 Feb to 15 Mar 2026): healthy, prediabetes and T2D; hypertension, dyslipidaemia, family history.
- EHR as FHIR R4 bundles: ICD-10, LOINC, ATC, TCF7L2, KCNQ1, SLC30A8 and 9p21 markers, DPDP style consent.
- Glucose from the UVA/Padova model with simglucose parameters and Dexcom noise, adapted for T2D. HR every minute, sleep stages, nightly HRV, steps.
- Apple Health and Google Fit export formats, parsed back by our adapters.
- One command regenerates everything in about 20 seconds from seed 20260315.



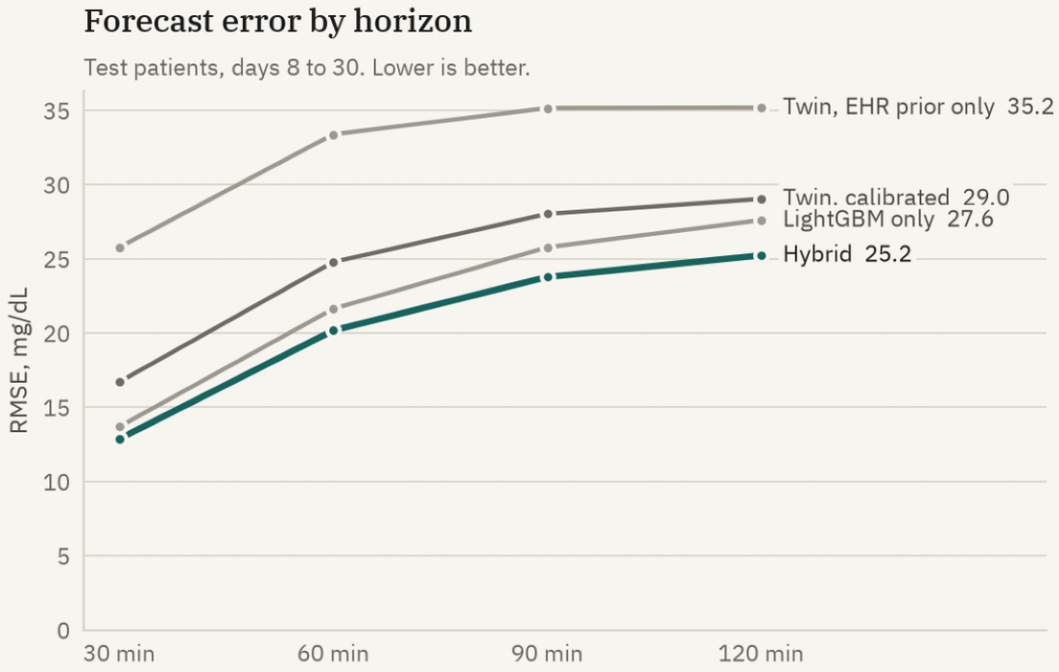
Fatima (synthetic): Ramadan pattern in her glucose profile.

METHOD

# Models

| Component | What it does |
|---|---|
| Twin ODE | dG/dt = -(SG + X)G + SG Gb H(t) + cf Ra(t) - eE E G, with insulin action, insulin, two gut compartments, exercise and metformin states |
| Calibration | MAP least squares on days 1 to 7 with EHR priors; about one second per patient |
| Residual model | LightGBM on twin error at 30, 60, 90, 120 min from CGM history, meals, activity, sleep, adherence, EHR |
| Uncertainty | CQR intervals at 90%; the alert tier uses the lower bound of the 50% interval, chosen on calibration patients |
| Cardio | Isolation forest and personal z scores (unsupervised) plus LightGBM, stacked by logistic regression |
| Explanations | Exact Shapley values over six twin inputs plus TreeSHAP on the residual, in mg/dL |

# Results: glucose forecasting

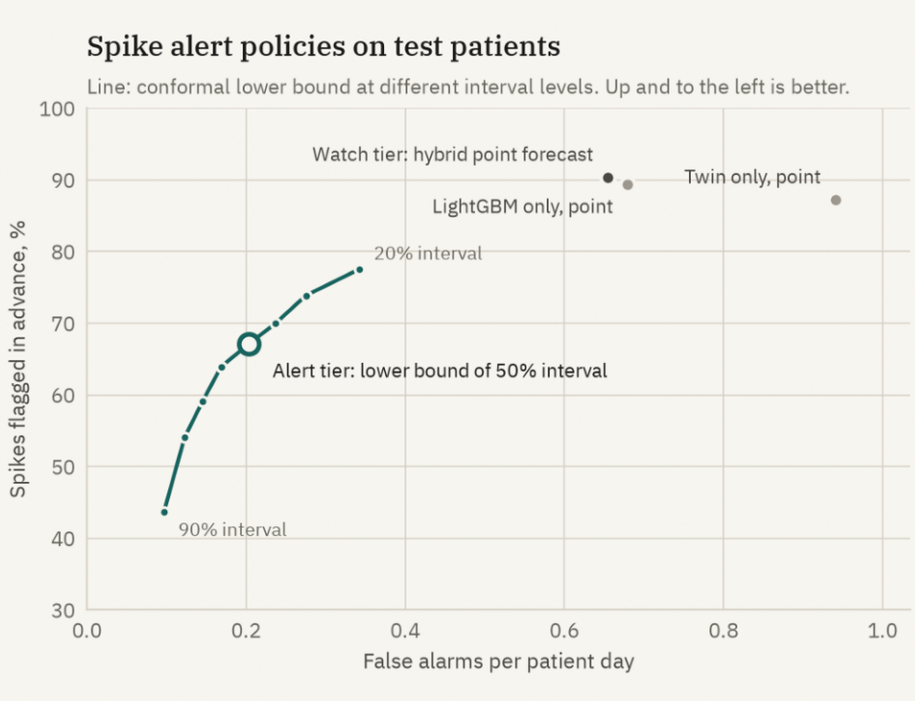


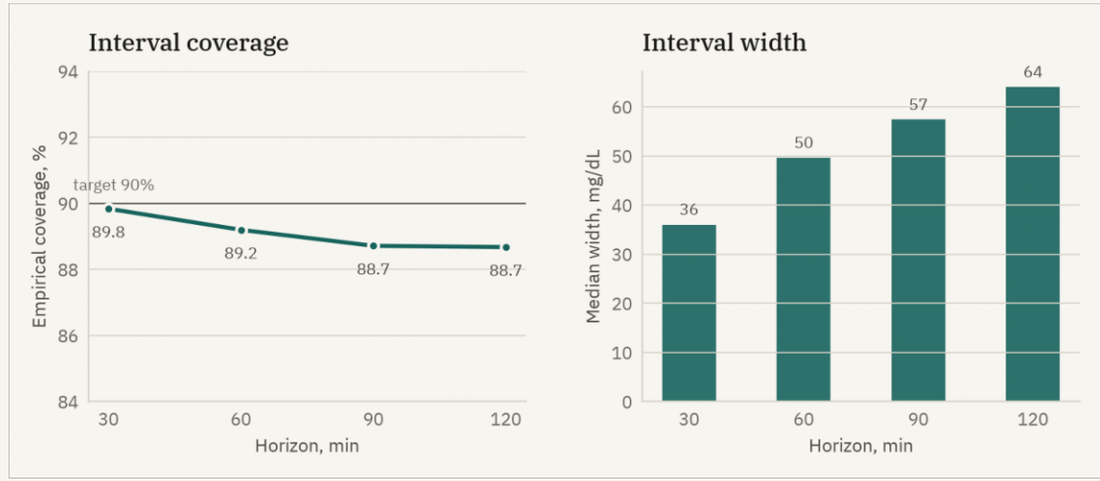

| RMSE, mg/dL | 30 | 60 | 120 |
|---|---|---|---|
| Twin, EHR only | 25.8 | 33.4 | 35.2 |
| Twin, calibrated | 16.7 | 24.8 | 29.0 |
| LightGBM only | 13.7 | 21.6 | 27.6 |
| **Hybrid** | **12.9** | **20.2** | **25.2** |

Hybrid beats both of its parts at every horizon. MARD at 60 min 11.1%. Clarke zone A 85.4% at 60 min, zones C to E 2.5%.

40 test patients never seen in training or calibration. Scored on days 8 to 30.

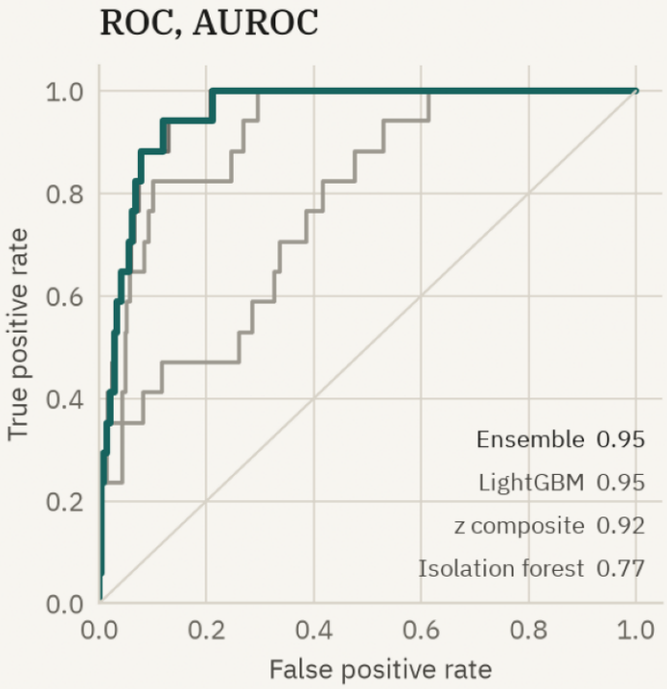
# Results: alerts and honest uncertainty

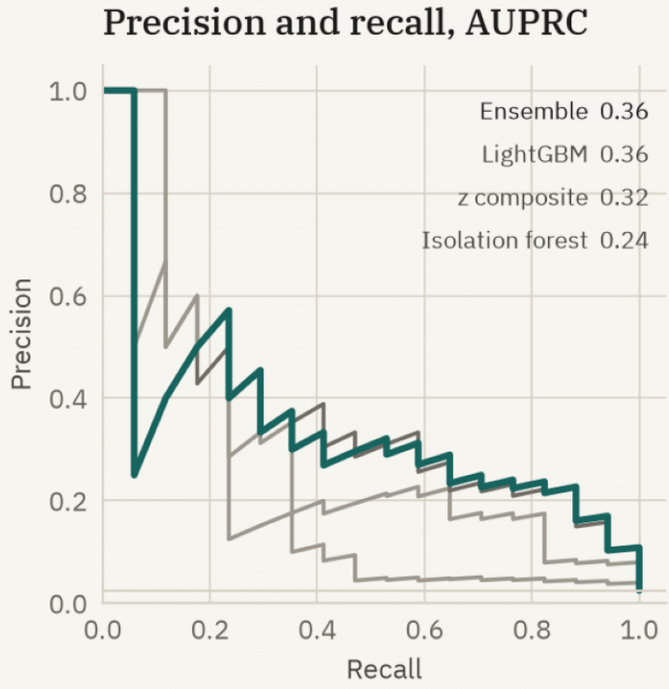


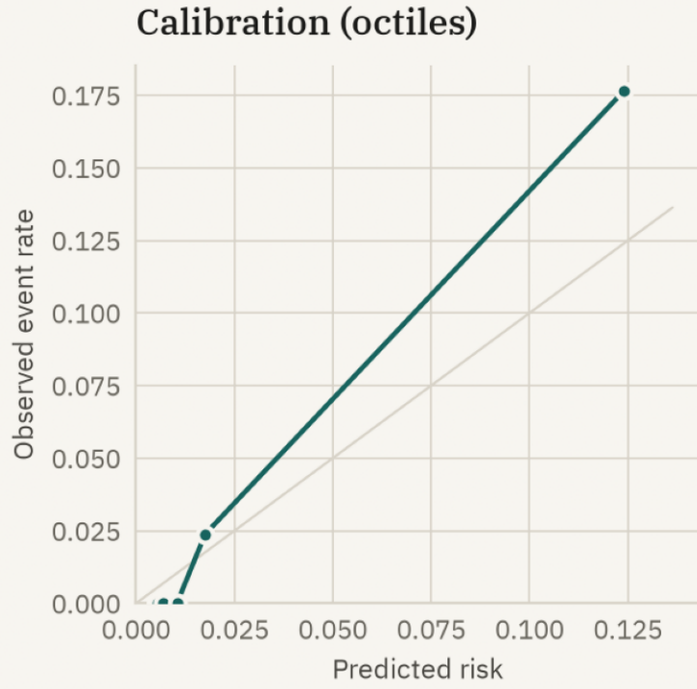




- Alert tier: 67% of spikes flagged, 85% precision, 0.20 false alarms per patient day, median 30 min ahead.
- Watch tier: 90% flagged a median 75 min ahead, shown but not pushed.
- Coverage of the 90% interval: 89.8% to 88.7% across horizons.

# Results: cardiovascular risk flag

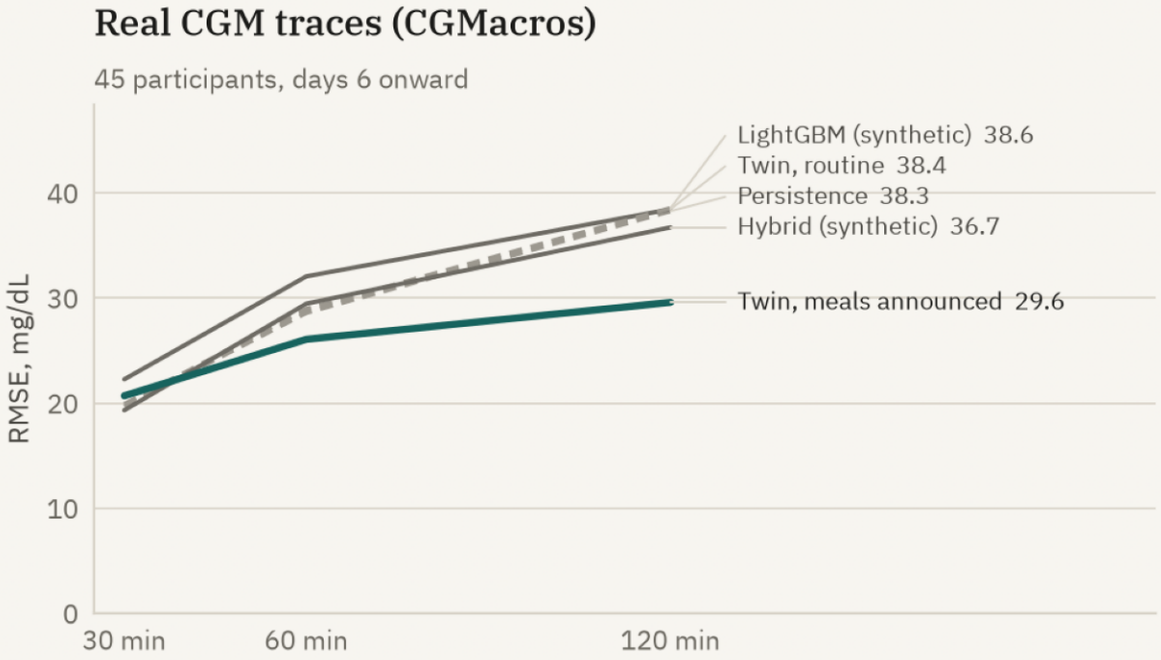


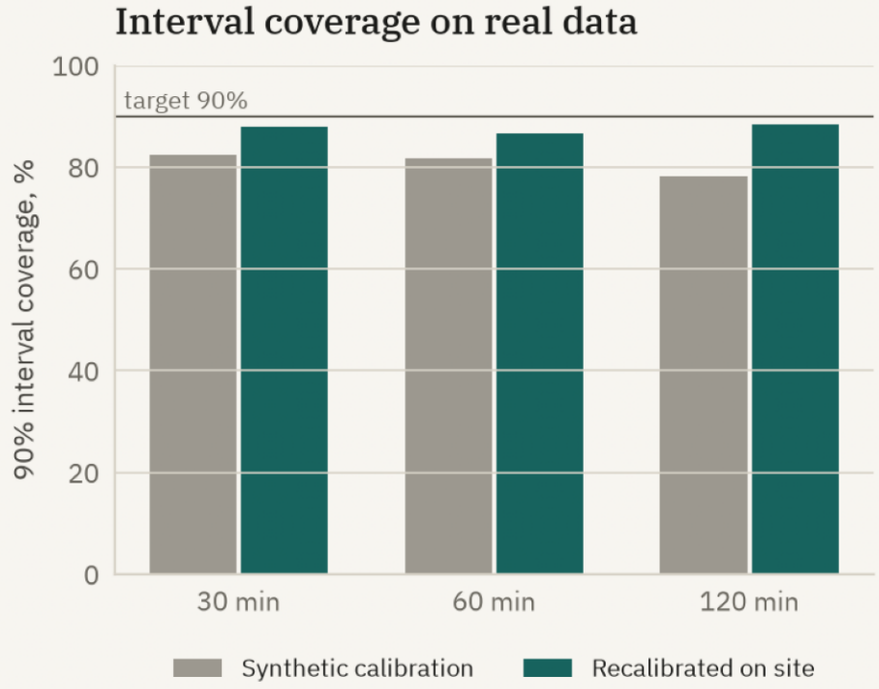




Ensemble AUROC 0.95 (95% CI 0.90 to 0.99), AUPRC 0.36 at 2.5% prevalence, sensitivity 88% at 90% specificity. At the 95% specificity operating point 5 of 6 test events were flagged up to 3 days ahead. Only 6 test events, so intervals are wide. The events come from our own simulator: with its warning signs weakened to half strength the flag catches 4 of 6, at a quarter 1 of 6. A demonstration of the pipeline, not a clinical claim.

# Real CGM traces: what transfers and what does not

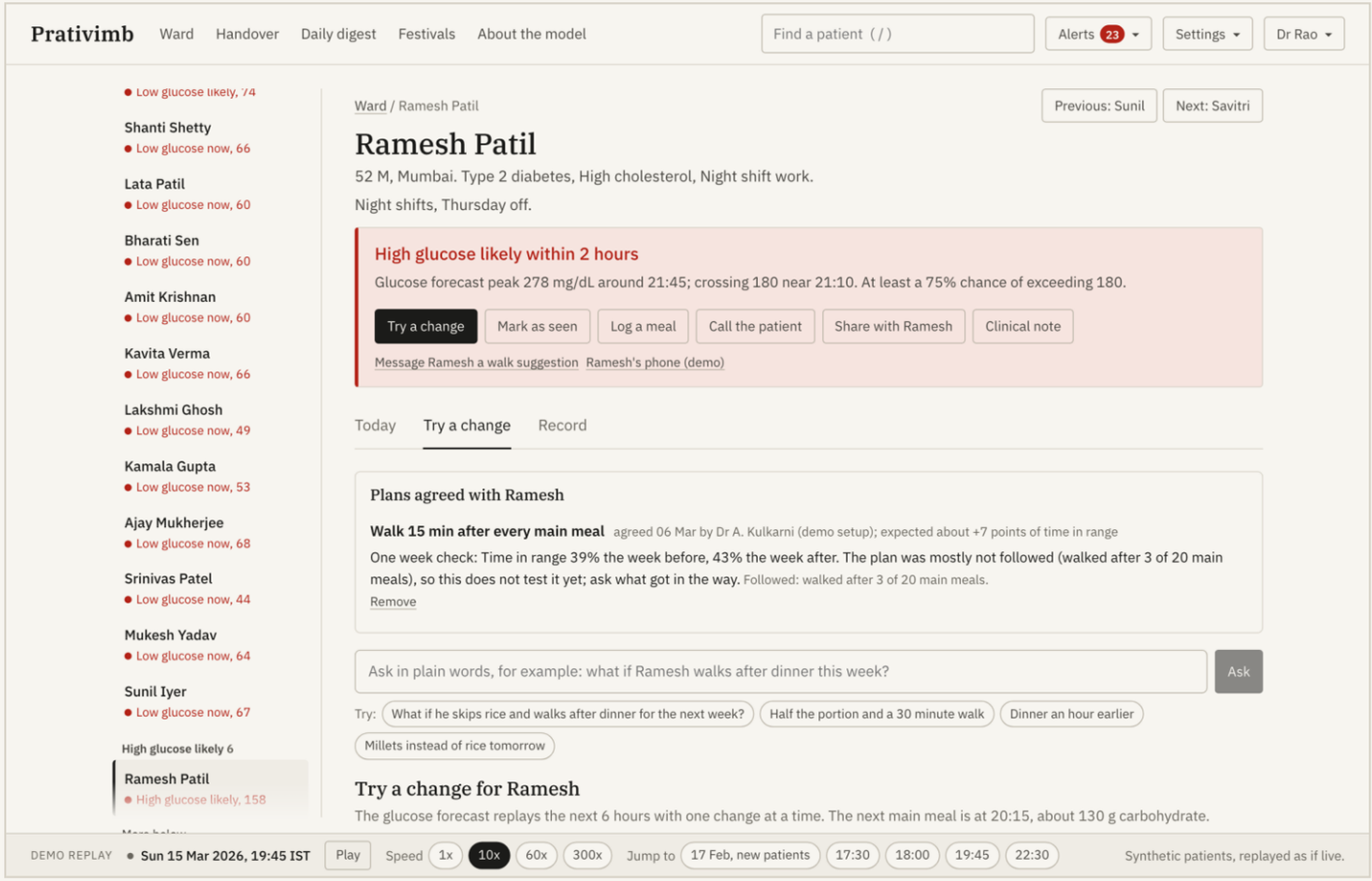



CGMacros (PhysioNet, open access): 45 adults, Dexcom G6, meals with known carbohydrate. Trained only on synthetic data, the hybrid lands near persistence (36.7 against 38.3 mg/dL at 2 h). Told about meals in advance, the calibrated twin reaches 29.6. Interval coverage recovers from 78% to 88% after per site recalibration.

CASE STUDY

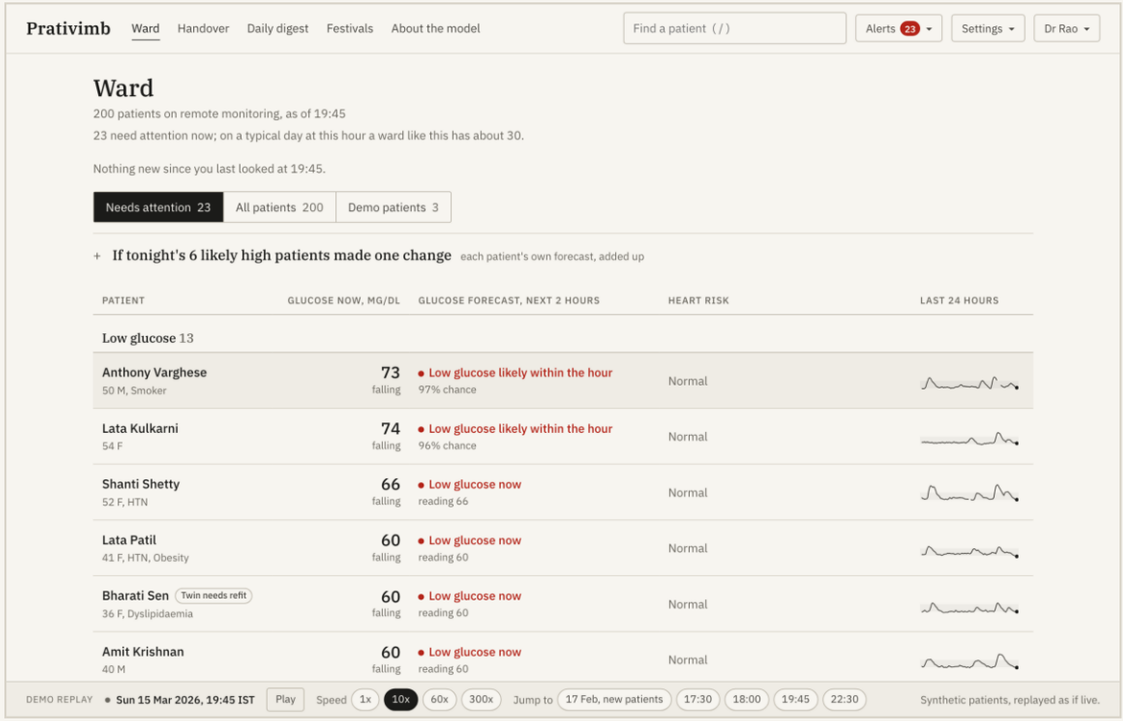
# What if: Ramesh, 52, night shift, T2D

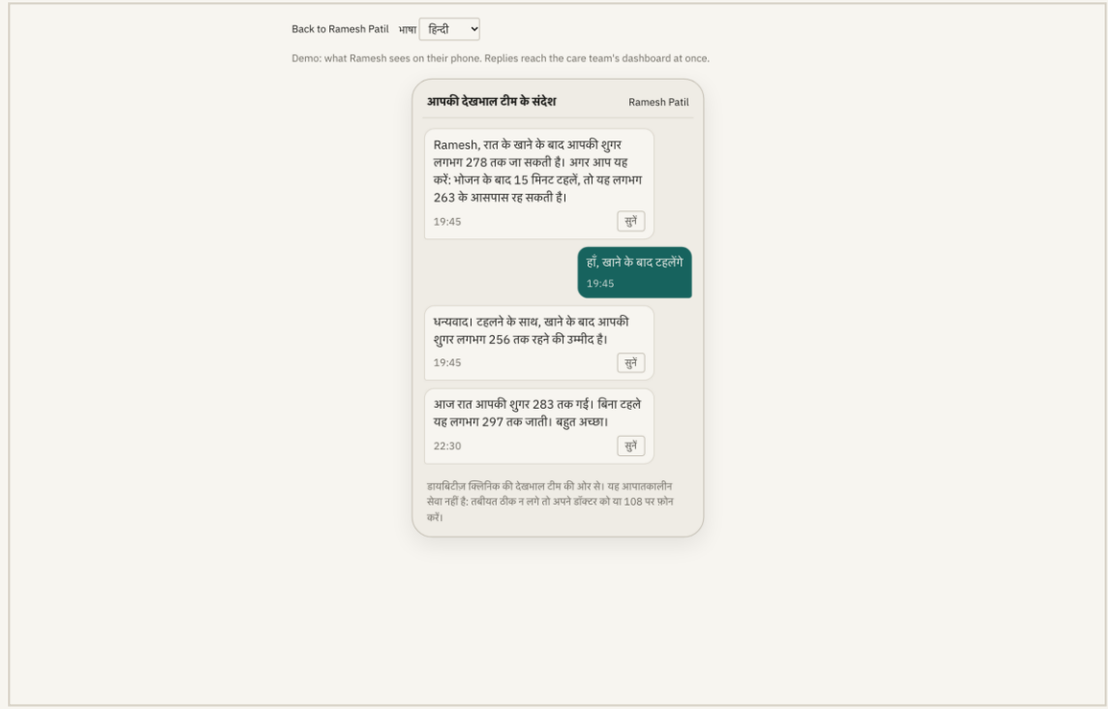



- 19:45. The hybrid forecast expects dinner at 20:15 and a peak near 278 mg/dL, after three nights of 4.9 h sleep.
- A 15 minute walk after dinner takes 15 mg/dL off the twin's 6 hour peak. Millets instead of white rice take 48.
- Together: peak 57 mg/dL lower and 40 minutes less above 180.
- Moving dinner earlier barely changes the peak tonight (it can only move to 19:50). The twin can say no.
- Checked against the reference simulator the twin never sees: across 316 moments in 40 test patients the twin gets the direction of a change right 76% of the time.

USE

# Clinical workflow




1. Ward grouped by urgency, with why each alert fired; n opens the next unseen alert. Load measured: 1.3 confident high alerts per patient per day, 94% came true. 2. The first action fits the alert: check now, call, book a review, try a change. Undo for 10 seconds.

3. A suggestion to the patient's phone in Hindi, Tamil or English; the reply enters the forecast and the outcome comes back to both sides. 4. Every alert's outcome, week plans checked a week later, handover, festival planner.

GOVERNANCE

# Privacy, safety and limits

- Synthetic data only: no personal data under the DPDP Act 2023 or HIPAA was used.
- Consent on screen: the patient can withdraw glucose, wearable or ABDM record sharing, and the service stops using that source.
- Roles checked by the service: doctor, nurse or health educator, patient. Every action and undo is audited.
- Deployment design: on premises inference, no identifiers in model features, clinic single sign on.
- Decision support only, not a diagnosis. The clinician decides.

## Limitations we state openly

- Synthetic ground truth is still a model; real data will be harder.
- Only 6 cardiovascular events in test patients, from our own simulator; no real outcome validation yet.
- Conformal coverage is approximate with serially correlated readings, and lower below 70 mg/dL.
- Unlogged meals remain the main source of missed spikes.

NEXT

# Impact and roadmap

| | |
|---|---|
| Now | Working prototype: data, twin, models, API, dashboard, tests, docs. Reproducible from one seed. |
| 3 months | Retrospective validation on de-identified CGM and EHR data under an ethics approval; Indian food absorption library. |
| 6 months | Federated learning of the residual model across hospitals; the twin stays per patient, only model updates move. |
| 9 months | ABDM integration (health ID, consent manager, FHIR); real device APIs (Libre, Dexcom, Health Connect). |
| 12 months | Prospective pilot in a diabetes clinic with alert burden and time in range as outcomes. |

*Impact: fewer hours above range, earlier review of cardiac drift, and conversations about change grounded in the patient's own physiology.*

Code: <public repository URL>.  Demo video: <unlisted YouTube URL>.  Thank you.